

ASSESSMENT TOOL 2 – MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 2 - Model Comparison Presentation (BERT vs GPT vs T5 vs LLaMA)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.</i> (BL3, BL4,BL5)		
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Section / Batch:	CSE AI	Date:	20/08/26

Instructions to Students

- Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.
 - Compare all four models: BERT, GPT, T5, and LLaMA.
 - Explain the **architecture and working principle** of each model.
 - Compare their **pre-training objectives and capabilities**.
 - Map each model to appropriate **real-world applications**.
 - Identify the **strengths and limitations** of each model.
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- Support the presentation using **credible research papers, official documentation, benchmarks, or other reliable sources**.

- Include at least one **comparison table or diagram**.
- Use relevant **real-world case studies or application examples**.
- Conclude by identifying the **most suitable model for the selected application** and justify the decision.
- Include proper **references/citations** in the presentation.

Code of Conduct

I A. Mounish / 192472273 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:

SECTION A : MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

APPLICATION MAPPING

18. Compare the four models for information extraction, including named entity recognition and relation extraction. Explain the advantages and limitations of each.

1. Question Requirement

Compare the four models for information extraction, including Named Entity Recognition (NER) and Relation Extraction. Explain the advantages and limitations of each.

2. Introduction

Information extraction converts unstructured text into useful structured information. Two important information-extraction tasks are Named Entity Recognition (NER), which identifies and classifies entities such as people, organizations and locations, and Relation Extraction, which identifies relationships between those entities.

BERT, GPT, T5 and LLaMA can all be applied to information extraction, but they use different Transformer architectures and therefore provide different strengths. BERT is strongly suited to contextual understanding, GPT and LLaMA support flexible instruction-based extraction, and T5 naturally represents extraction as a text-to-text transformation.

3. Named Entity Recognition (NER)

NER identifies important entities in a sentence and assigns them categories. Common categories include PERSON, ORGANIZATION, LOCATION, DATE and MONEY.

Example: “Elon Musk founded SpaceX in the United States.” A system can identify Elon Musk as a PERSON, SpaceX as an ORGANIZATION, and United States as a LOCATION.

4. Relation Extraction

Relation Extraction identifies the relationship between two or more entities. From the sentence “Elon Musk founded SpaceX,” an extraction system can produce the relation “Elon Musk → founded → SpaceX.”

This converts information expressed naturally in a sentence into a structured form that can be used in knowledge graphs, search systems, question answering and other NLP applications.

5. BERT for Information Extraction

5.1 Information Extraction

BERT is highly suitable for information extraction because its bidirectional self-attention provides strong contextual representations. It can use information from both the left and right sides of a word, which helps determine the role and meaning of entities in context.

5.2 NER

BERT can be fine-tuned for token-level NER. For example, in “Elon Musk founded SpaceX in the United States,” a BERT-based NER system can identify Elon Musk as PERSON, SpaceX as ORGANIZATION, and United States as LOCATION.

5.3 Relation Extraction

BERT can also be adapted to relation extraction. For “Elon Musk founded SpaceX,” the system can identify the subject as Elon Musk, the relation as founded, and the object as SpaceX.

5.4 Advantages

- Strong bidirectional contextual understanding.
- Excellent suitability for NER and token-level classification.
- Can be fine-tuned for specific extraction domains.
- Produces relatively predictable task-specific outputs.

5.5 Limitations

- Usually requires task-specific fine-tuning or an additional extraction layer.
- Not primarily designed for free-form text generation.
- Complex relation extraction may require task-specific architecture.

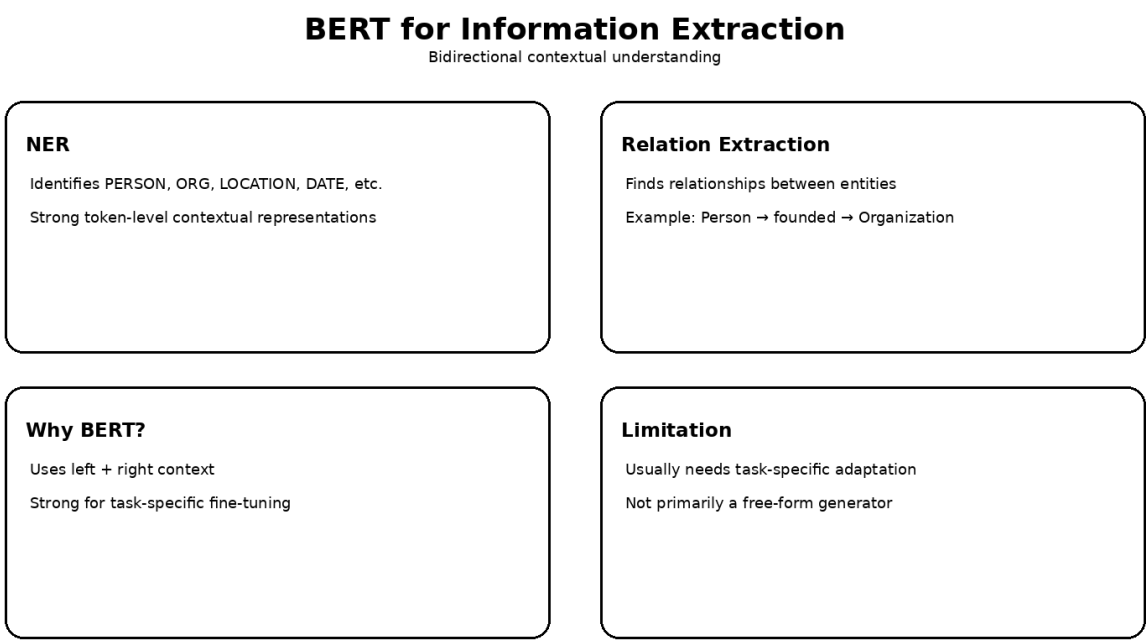


Figure 1. BERT information-extraction visual: NER, relation extraction, contextual advantage and limitation.

6. GPT for Information Extraction

6.1 Information Extraction

GPT can perform information extraction through prompting and instruction-based methods. Instead of requiring a separate classifier for every extraction format, the user can describe the desired entities and relationships in natural language.

6.2 NER

For NER, GPT can be instructed to extract people, organizations and locations from a paragraph. It can return the results as a list, table or structured format.

6.3 Relation Extraction

GPT can identify subject–relation–object patterns. For example, from “Microsoft acquired GitHub,” it can extract Microsoft as the subject, acquired as the relation, and GitHub as the object.

6.4 Advantages

- Strong natural-language understanding.
- Flexible instruction-based extraction.
- Can handle different entity and relation types.
- Can combine extraction with summarization or question answering.
- Can produce structured output when appropriately instructed.

6.5 Limitations

- Results can vary with prompt design.
- Generated information may not always be explicitly present in the source.
- Output should be validated for accuracy and consistency.
- Large models may require significant computational resources.

GPT for Information Extraction

Instruction-based and generative extraction

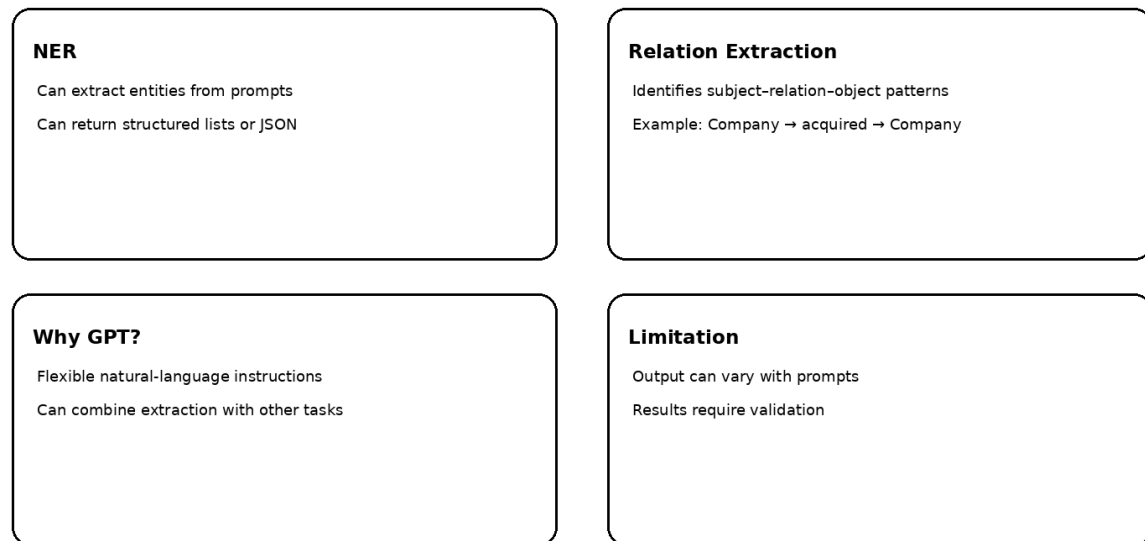


Figure 2. GPT information-extraction visual: instruction-based NER and relation extraction.

7. T5 for Information Extraction

7.1 Information Extraction

T5 uses an encoder-decoder Transformer architecture and treats NLP tasks as text-to-text problems. Information extraction can therefore be expressed as converting a source sentence into a sequence containing entities and relationships.

7.2 NER

T5 can receive a sentence and generate the detected entities with their categories. For example, “Elon Musk founded SpaceX” can be transformed into an output containing “Elon Musk – PERSON” and “SpaceX – ORGANIZATION.”

7.3 Relation Extraction

T5 can transform a sentence into a structured relationship such as “Elon Musk → founded → SpaceX.” Its encoder represents the input and its decoder generates the required extraction result.

7.4 Advantages

- Naturally supports text-to-text information extraction.
- Encoder provides strong input understanding.
- Decoder can generate the desired extraction format.
- Flexible for multiple extraction tasks.

7.5 Limitations

- Output format must be clearly controlled.
- Generated extraction may require validation.
- Encoder-decoder processing is more involved.
- Task-specific training or adaptation may be needed for high-quality extraction.

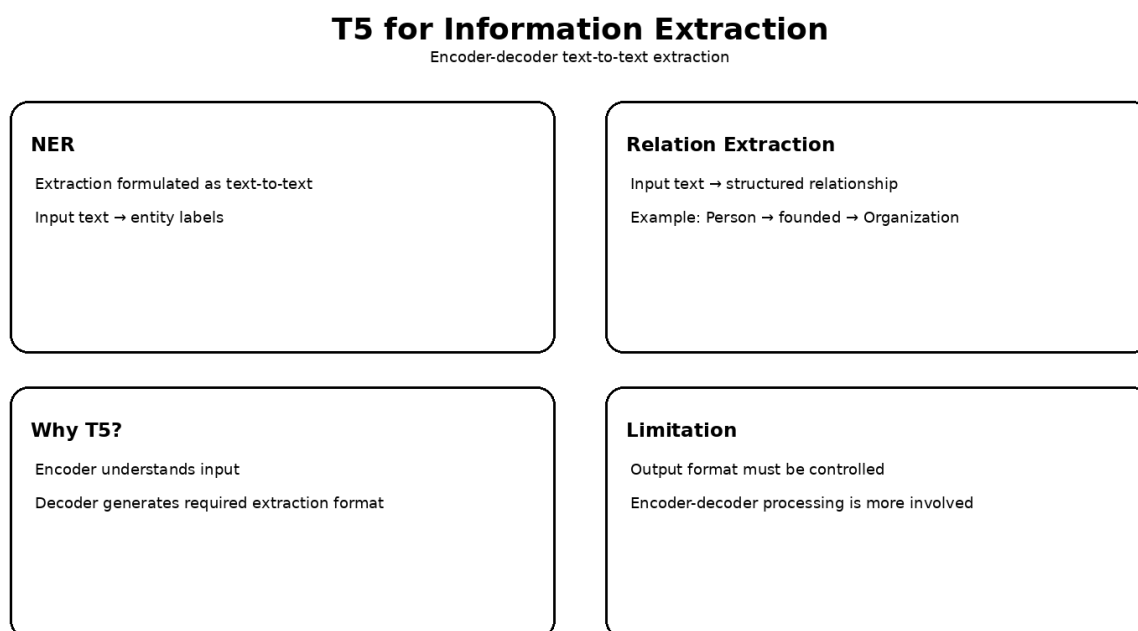


Figure 3. T5 information-extraction visual: text-to-text NER and relation extraction.

8. LLaMA for Information Extraction

8.1 Information Extraction

LLaMA can perform information extraction using contextual language understanding and instruction-based generation. It can be prompted to identify entities and relationships from unstructured text.

8.2 NER

LLaMA can identify multiple entity categories from a document. For example, it can extract Elon Musk as PERSON, SpaceX as ORGANIZATION and United States as LOCATION from an appropriate input.

8.3 Relation Extraction

LLaMA can identify relationships between entities. From “Microsoft acquired GitHub,” it can extract Microsoft → acquired → GitHub.

8.4 Contextual Example

In a multi-sentence passage, LLaMA can use surrounding context to interpret references such as “the company” and connect them to an earlier entity when the context supports that interpretation.

8.5 Advantages

- Strong contextual language capability.
- Flexible prompt-based extraction.
- Can handle multiple entity and relationship types.
- Can combine extraction with summarization and question answering.
- Useful for general-purpose information extraction systems.

8.6 Limitations

- Generated results can sometimes be inaccurate.
- Prompt design can influence extraction quality.
- Output should be verified for factual consistency.
- Large variants may require substantial memory and computation.

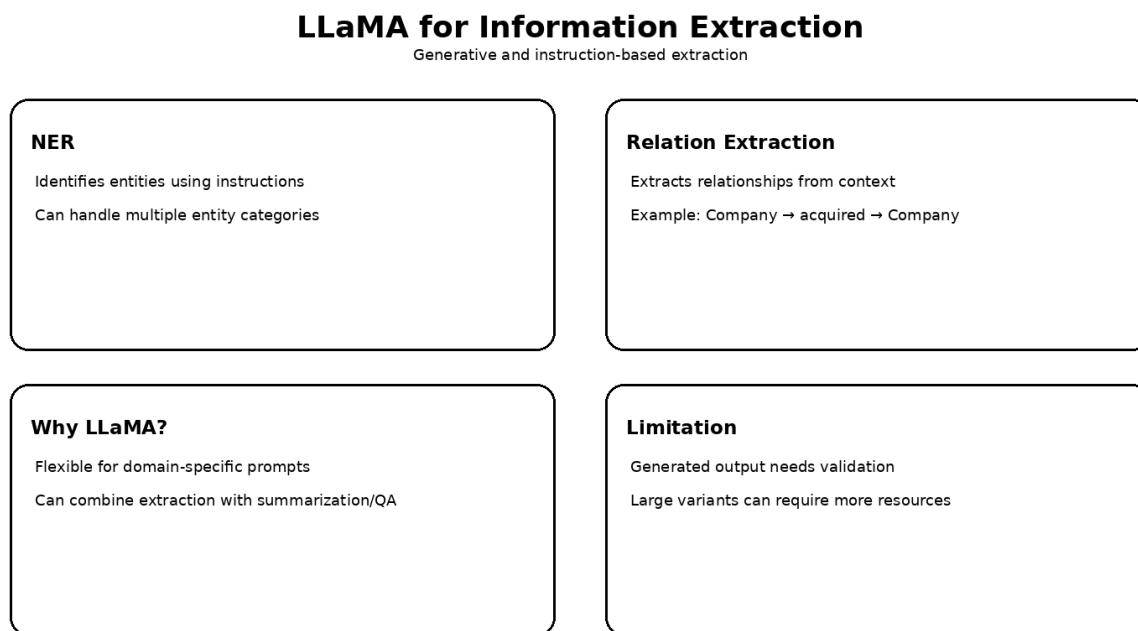


Figure 4. LLaMA information-extraction visual: instruction-based NER, relation extraction and limitations.

9. Comparison of BERT, GPT, T5 and LLaMA

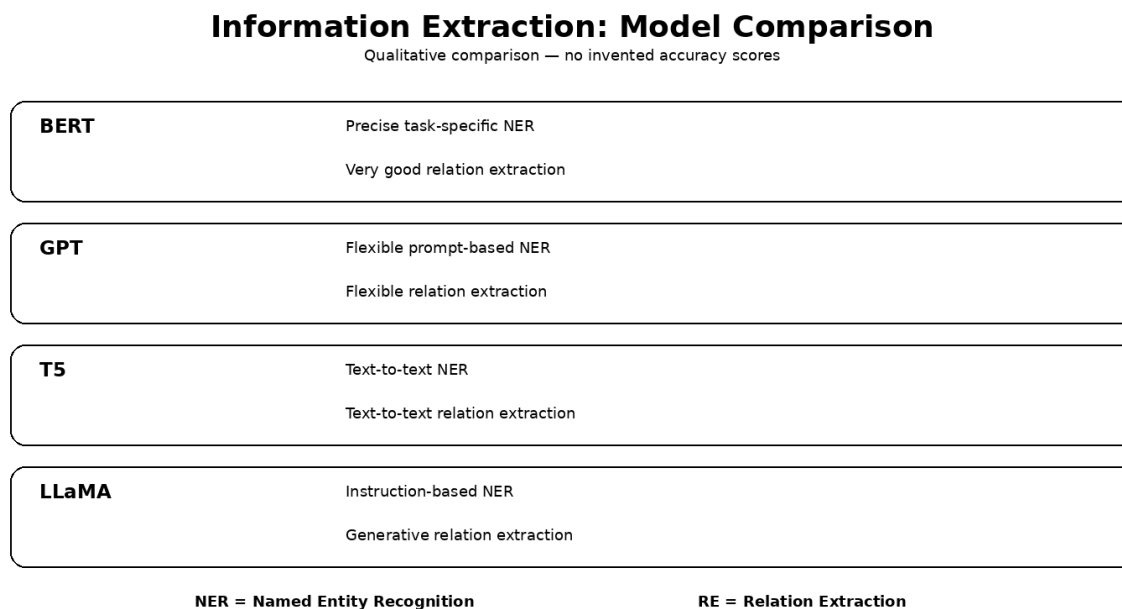


Figure 5. Qualitative comparison of the four models for NER and Relation Extraction.

Feature	BERT	GPT	T5	LLaMA	Best Characteristic
Architecture	Encoder-only	Decoder-only	Encoder-decoder	Decoder-only	—
NER	Excellent	Good	Very Good	Good	BERT
Relation Extraction	Very Good	Good	Very Good	Good	BERT / T5
Context	Bidirectional	Preceding context	Full encoded input + decoder context	Preceding context	BERT / T5
Extraction style	Fine-tuned classification	Prompt / generation	Text-to-text	Prompt / generation	Depends on task
Output flexibility	Good	Very High	Very High	Very High	GPT / T5 / LLaMA
Main limitation	Task-specific adaptation	Validation needed	Format control	Validation needed	—

10. Advantages and Limitations – Direct Comparison

BERT

Best suited when precise contextual understanding and task-specific NER are the priority. Its main limitation is the need for task-specific adaptation.

GPT

Best suited when extraction must be flexible and instruction-driven. Its main limitation is that generated outputs can vary and require validation.

T5

Best suited when extraction can be formulated as a text-to-text transformation. Its main limitation is the need to control the generated output format.

LLaMA

Best suited for flexible generative extraction and systems that combine extraction with other language tasks. Its main limitation is the need to validate generated information.

11. Final Conclusion

BERT, GPT, T5 and LLaMA can all support information extraction, but their architectures lead to different strengths. BERT is particularly strong for precise, task-specific NER and contextual extraction. GPT provides flexible instruction-based extraction. T5 naturally handles extraction as a text-to-text transformation. LLaMA provides flexible generative extraction and can combine extraction with other language tasks.

For high-precision NER and controlled extraction, BERT is a strong choice. For flexible prompt-based extraction, GPT or LLaMA can be useful. For structured text-to-text extraction, T5 is particularly appropriate. The final selection should depend on the required precision, output format, flexibility and computational constraints.